Web Service Recommendation via Combining Topic-Aware Heterogeneous Graph Representation and Interactive Semantic Enhancement

Buqing Cao, Qian Peng, Xiang Xie, Zhenlian Peng, Jianxun Liu, and Zibin Zheng, *Fellow, IEEE*

***Abstract*—With the continually increasing number of Web services, it becomes a challenging task to efficiently and accurately provide Web services that meet developers' functional requirements. Existing heterogeneous graph-based service recommendation methods simply utilize the heterogeneous structural features of the service network and suffer from the missing and blurring of service interaction semantic information due to the characteristics of meta-paths. In fact, service node description documents contain fine-grained semantics generated by multifaceted topic-aware factors, but few efforts are committed to mining them. Therefore, a Web service recommendation method via combining topic-aware heterogeneous graph representation and interactive semantic enhancement is proposed in this paper. It employs an alternating two-step aggregation mechanism, including meta-path instance intra-decomposition and meta-path inter-integration, which uniquely aggregates topic-aware factors according to the inferred topic distributions while preserving structural semantics. Additionally, it introduces the topic prior knowledge guidance module to improve the quality of the inference's topic factors. Simultaneously, the method designs the interactive semantic enhancement module to address the missing and blurring of service interaction semantic information caused by meta-paths. The module explores complex interaction patterns among services and utilizes personalized knowledge meta-network to enhance contrastive learning of service interaction semantics, allowing the personalized knowledge transformer with adaptive contrastive enhancement. The experimental results on the real dataset of ProgrammableWeb show that compared with the other nine methods, the proposed method has better service recommendation performance on evaluation metrics HR and NDCG representing accuracy and satisfaction, respectively.**

***Index Terms*—Service recommendation, topic-aware, meta-paths, attention mechanism, meta-network, contrastive learning.**

I. Introduction

With the popularity of cloud computing and service computing, the number of Web APIs continues to grow. The emergence of a plethora of Web services poses difficulties for developers in Web service selection and discovery. Web service recommendation can assist developers in identifying the most suitable services from the huge library of Web APIs, alleviating their burden and promoting the reuse and sharing of Web services, thereby accelerating application development. Therefore, to address developers' difficulties in Web service selection and discovery as well as to facilitate the creation of new service compositions (Mashups), service recommendation becomes crucial.

Currently, the research methods for Web service recommendation mainly include functional feature-based service recommendation methods, non-functional feature-based service recommendation methods, and hybrid feature-based service recommendation methods according to the type of features. Traditional functional feature-based service recommendation methods mainly extract the key features of Web services from the functional description documents of Web Service Description Language (WSDL) and calculate the similarity between services using similarity functions [1]. However, due to the limited length and unstructured characteristics of the service description document, it is difficult to obtain accurate and complete service features, which leads to unsatisfactory results in Web service recommendation. Most of service recommendation methods based on non-functional features use Quality of Service (QoS) and historical interaction information as the basis for service recommendation [2], which consider multidimensional evaluation criteria, are not only limited to functional matching, but also can provide a more comprehensive service selection. However, non-functional QoS information (such as stability, security, timeliness, etc.) is difficult to be acquired [3]. In addition, non-functional feature-based service recommendation methods suffer from matrix sparsity and cold-start problems due to the increase in the number of Web services, and have limited recommendation capability for new developers, and can only learn linear interactions between services.

Many researchers adopt the hybrid feature methods that combine functional and non-functional features of services to improve the accuracy of service recommendation. These methods comprehensively consider the functional and non-functional aspects of services and can better solve the problems of data sparsity and incompleteness, thereby providing more comprehensive and personalized service recommendation results. For example, some researchers have introduced auxiliary

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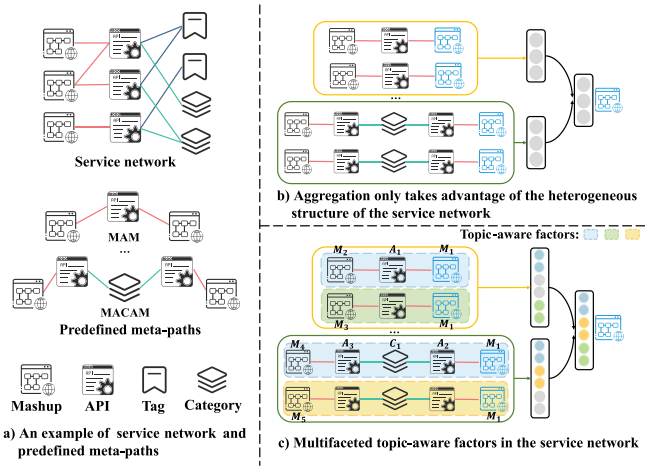

Fig. 1. An illustration of service network and structural comparison between our method and previous methods.

information (such as tag, structural information, etc.) during the training process of topic models to extract more important and accurate latent semantic information of Web services [4], [5], [6]. Other researchers try to construct service information into a service network consisting of various services and their relationships. Due to the sparsity and complexity of service networks, it is challenging to effectively model all aspects of service semantics. Considering that Graph Neural Networks (GNNs) can more accurately represent and learn relational information between nodes in complex network structures, many scholars have carried out a lot of related work on service networks [7], [8], [9], [10]. Although these methods enable service nodes to learn information about neighboring services in the service network, they do not consider the heterogeneity of Web service networks, which leads to the loss of unique information and rich semantics in the service network. To address this problem, Heterogeneous Graph Neural Networks (HGNNs) have recently been widely used in many fields [11], [12], [13], since HGNNs can define and utilize meta-paths to reveal rich semantic and structural information in heterogeneous structures. For example, in the service network shown in Fig. 1(a), two predefined meta-paths (MAM and MACAM) can reveal two different semantics of the Mashup node, i.e., calling the same APIs and calling the same category of APIs. A classical paradigm for applying HGNNs to service networks is to adopt a hierarchical aggregation on the semantic level of the service nodes and the meta-paths in order to integrate semantic information from different predefined meta-paths semantic [13], as shown in Fig. 1(b).

Although HGNN-based methods can reveal rich semantics of different aspects of service nodes, they are limited to utilizing only structural features when dealing with highly complex service networks. Unstructured text content in service nodes contains potential semantics generated by multifaceted topic-aware factors, and these potential topic-aware semantics are more fine-grained than structural semantics. For example, the meta-path instance-based context can reflect that the Mashup M_1 in Fig. 1(c) is interested in multiple APIs with different topics. In the MAM instance-based context, Mashup M_1 and Mashup M_2 both call API A_1 because they have shared interest in the topic "Traffic" (shown in blue), Mashup M_1 and Mashup M_3 jointly call the API with the topic "Search" (shown in green). Similarly, in the MAM instance-based context, both Mashup M_1 and Mashup M_4 call the API for the map category (shown in C_1), but only include the more fine-grained "Traffic" (shown in blue) topic in the C_1 category, and both Mashup M_1 and Mashup M_5 call the API included under the topic "Online" (shown in yellow). Therefore, it is essential to consider such fine-grained topic-aware semantic information of service nodes rather than simply fusing confused features, ignoring this information will inevitably impact the accuracy of service recommendations [14].

In our preliminary study [15], Topic-aware Heterogeneous Graph Neural Network (THGNN) is applied to combine the structural semantics and topic-aware semantics of service network to improve the accuracy of Web service link prediction. Service link prediction serves as a crucial foundation for service recommendation. Extensive experiments show that the proposed service link prediction method achieves better prediction results than the baseline method. Thus, the previous study demonstrated that THGNN can capture finer-grained topic-aware semantics in service networks. The structural semantics of THGNN are captured by meta-paths. This structural semantics is more like the similarity structural semantics of the nodes at both ends of the meta-paths. Meta-paths are also the current common method to capture the structural semantics of heterogeneous graphs. However, due to the specific path pattern and indirectness of meta-paths, the interactive semantic information between Mashups and APIs is lost and blurred, but numerous studies show that this interactive information is very important for recommendations [16], [17], [18]. To this end, a Web service recommendation method via combining topic-aware heterogeneous graph representation and interactive semantic enhancement, named THISE is proposed in this paper, which introduces interactive semantic enhancement based on the topic-aware heterogeneous graph neural network. Firstly, the method extracts prior topic distributions from Web service description documents using the LDA topic model [19], and then it utilizes service nodes and auxiliary information nodes (tags and categories) to construct a service network. Subsequently, it employs three modules of the topic-aware heterogeneous graph neural network to extract and integrate fine-grained topic-aware semantics. The three modules include meta-path instance intra-decomposition, meta-path inter-integration, and topic prior knowledge guidance. Meta-path instance intra-decomposition aims to infer the topic-aware distribution of the meta-path instance-based context and aggregate context information according to this distribution, thereby forming multifaceted representations and obtaining fine-grained topic-aware semantics. Meta-path inter-integration employs a multi-facet attention mechanism to integrate different meta-paths, combining both similarity structure semantics and fine-grained topic-aware semantics. Topic prior knowledge guidance utilizes the prior topic distributions as global knowledge to help guide the aggregation of context information, making the inferred topic-aware subspace more orthogonal and enhancing the interpretability of topic-aware representations. Finally,

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interactive semantic enhancement is utilized to address the problem of missing and blurring interactive semantic information caused by meta-paths. The interactive semantic enhancement module captures the interaction semantics between Mashups and APIs from the complex interaction patterns among services, and utilizes the personalized knowledge meta-network to enhance the contrastive learning of service interaction semantics, so as to enable the transfer of personalized knowledge with adaptive contrastive enhancement, and to generate more robust feature representations for service nodes.

In summary, the contributions of this paper are as below:

- A topic-aware heterogeneous graph neural network-based Web service representation and fine-grained topic-aware semantic extraction approach is presented. The approach effectively aggregates diverse and complex semantic information in the service network based on the inferred multifaceted topic-aware distributions so that the service nodes retain both structural semantics and topic-aware semantics. In addition, it uses global prior knowledge extracted from service node description documents to guide the topic-aware factor inference process, improving the quality of topic-aware semantics.
- A novel Web service recommendation method is designed. the method introduces interaction-aware semantic mining and personalized knowledge meta-network based on Web service representation that combines similarity structural semantics and topic-aware semantics, and it uses contrastive learning for adaptive enhancement of interactive semantics to strengthen interactive semantics that are very important for Web service recommendation, and to improve the accuracy of Web service recommendation.
- Experimental results on the real dataset ProgrammableWeb show that the method proposed in this paper outperforms other existing baseline methods on the Web service recommendation task, which successfully validates the effectiveness of the method proposed in this paper.

The rest of this paper is organized as follows: Section II is related work. Section III introduces relevant definitions and terminology. Section IV details the method proposed in this paper. Section V carries out the concrete experiment process and analysis. Finally, conclusions and future work are presented in Section VI.

II. Related Works

Existing Web service recommendation methods can be divided into three categories based on service features, i.e., functional features-based service recommendation, non-functional features-based service recommendation, and hybrid features-based service recommendation.

A. Functional Features-Based Service Recommendation

The main idea of functional features-based service recommendation is to analyze the functional features in the developer's requirement document and the Web service description document. It calculates the similarity between them and forms a recommendation list of services that match the developer's requirement with higher similarity to the developer. In previous research works, word embedding models are usually used to compute the similarity between Web services. For example, Lei et al. [20] utilized the TF-IDF weighting method [21] to compute the service similarity between the semantic information in the service description document and the topic and functionality of the service, so as to recommend appropriate component services for composite services. Wu et al. [22] proposed a mixed Web service recommendation method WSTRec, which uses tag co-occurrence, tag mining, and semantic relevance metrics for tag recommendation to facilitate service discovery. Wu et al. [23] presented DivMTID, a service recommendation method with high accuracy and diversity, which utilizes users' historical ratings of Web services to explore users' preferences and computes a weight vector for each Web service using the TF-IDF algorithm. Next, cosine similarity is utilized to compute the similarity between candidate Web services and historical Web services, and then the ranking scores of candidate Web services are predicted and a diverse list of service recommendations is generated. In addition to using word embedding models, probabilistic topic models are also widely used as text similarity methods for Web service recommendation. Li et al. [24] proposed a service recommendation probabilistic model that uses the LDA topic model [19] to represent the relationship between Mashups, APIs, and their links. Zhang et al. [25] used a time series LDA model to extract the user's dynamic preferences, combined with an LSTM neural network to predict the next stage of preferences based on the dynamic sequence of user preferences and recommended services to the user according to the predicted preferences. Chen et al. [26] raised a Web service recommendation method that combines word embedding and topic modeling by first constructing a high-quality word embedding model using English Wikipedia, and then using the GPU-DMM model to model short texts to obtain their implied topic distributions and calculating service similarity. Methods based on text similarity or topic similarity usually require mining the functional description documents of Web services. However, the challenge is that the functional description documents of many Web services are either missing or have very limited content. The limitations of functional description documents make it difficult to obtain valid service representations, which affects the accuracy and effectiveness of Web service recommendations.

B. Non-Functional Features-Based Service Recommendation

The core concept of non-functional features-based service recommendation is to extract user preference characteristics from historical service usage data, such as response time, reliability, availability, and popularity, which represent Quality of Service (QoS) information [27]. Subsequently, algorithms like Collaborative Filtering (CF) and Matrix Factorization (MF) are employed to predict missing interaction data between users and services, facilitating the recommendation of services. Chen et al. [28] used location information and QoS values to cluster users and services and personalized service recommendations to users based on the clustering results. Hu et al. [29] proposed an improved time-aware collaborative filtering method for Web

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service recommendation by integrating temporal information into similarity measurement and QoS prediction. Ren et al. [30] obtained a separating hyperplane from the historical rating data so as to filter the services that the user may dislike based on Support Vector Machines (SVM) and CF method, in addition to measuring the degree of preference of the user by measuring the distance between the point representing the service and the separating hyperplane, and ultimately recommending the service based on the predicted degree of preference. Chattopadhyay et al. [31] constructed the user-intensive model and the service-intensive model through collaborative filtering to generate a set of similar users on a set of similar services, and finally used a neural network-based regression model to predict the QoS values of the target users invoking the target services. Wang et al. [32] proposed a method to construct data with high-dimensional temporal and spatial information into a reliable multidimensional service quality dataset, which improves the quality of Web service recommendations through tensor decomposition and reconstruction optimization algorithms. Cheng et al. [33] designed QoSHTD, a Web service recommendation method based on QoS prediction and hierarchical tensor decomposition. Kang et al. [34] proposed a novel neural network architecture called NAFM to address the drawbacks of Factorization Machine (FM) ignoring nonlinearities and complex inherent structures in the data. The architecture utilizes neural networks for nonlinear feature interactions and uses an attention network for obtaining the weights of the interactions between the nodes in order to improve the recommendation accuracy. Chang et al. [35] proposed a new graph-based matrix decomposition method for QoS prediction, which integrates the multifaceted relationships of user-perceived context and service-perceived context and mines deeply into the potential relationships based on the QoS matrix. However, with the increasing number of users and Web services, Web service recommendation methods represented by collaborative filtering and matrix decomposition suffer from data sparsity as well as vector representation of high dimensionality, which limits the improvement of service recommendation performance.

C. Hybrid Features-Based Service Recommendation

By comprehensively considering the functional and non-functional features of services, the association relationship between services can be more comprehensively represented, thus improving the accuracy of service recommendation, and hybrid features-based service recommendation has become a current research hotspot. Yao et al. [36] proposed a unified CF and content-based Web service recommendation method that uses a probabilistic generative model to synthesize QoS information and semantic functional data of Web services. Cao et al. [37] developed a two-level topic model to improve the accuracy of service clustering, and based on the clustering results, designed a CF-based Web API recommendation method. Botangen et al. [38] presented a geographical-aware collaborative filtering service recommendation method that combines geographic information with functional descriptions and adds this information to a matrix decomposition model to improve the accuracy of Web API recommendations. Xiao et al. [39] raised a hybrid deep learning approach combining CF and textual content to integrate the call interactions between Mashups and services as well as their functional features into a deep neural network to improve the performance of service recommendations. Cao et al. [3] proposed a bilinear neural network and Extreme Deep Factorization Machine (xDeepFM) quality prediction-based Web service recommendation, which is based on content-oriented and structure-oriented classification of service functionality, and is combined with multi-dimensional quality attribute-based service call prediction. Cao et al. [2] presented a Web service recommendation method integrating heterogeneous graph attention network representation and Feature Importance and Bilinear feature Interaction Network (FiBiNET) score prediction, which first defines meta-paths on a heterogeneous information service network based on different semantic information, constructs service similarity matrices using exchangeable matrices and meta-path-based similarity metrics, and designs a two-layer attention model to derive feature representations of Web services, and, finally, dynamically learns the significance of feature and complex feature interactions and predicts the score of Web services using FiBiNET. Similarly, the method proposed in this paper is a hybrid features-based service recommendation method. Different from previous work that only considers a single topic semantics or a single heterogeneous structural semantics, the method in this paper captures the potential and finer-grained topic semantics carried in the unstructured textual content of the service nodes generated by multiple topic-aware factors, fully utilizes the heterogeneous structure of the service network and the unstructured text content. The method combines structural semantics with finer-grained multifaceted topic-aware semantics to generate service node representations that preserve both structural and multifaceted topic-aware semantics. In addition, based on retaining the structural semantics and multifaceted topic-aware semantics of service representations, the method designs an interactive semantics enhancement module for solving the problem of missing and blurring interaction semantics information due to meta-paths, and generates more robust feature representations for service nodes to improve the quality of Web service recommendation.

III. Preliminaries

The preliminaries of the proposed method are presented, including the related concepts of Web service network, meta-path instance, and meta-path instance-based context.

Definition 1. Web Service Network

Web service networks are often heterogeneous, containing interactive semantics from Mashups and APIs, as well as auxiliary information from tags, categories, and description documents. Web service network can be formalized as $SNet = \{\mathcal{V}_m, \mathcal{V}_a, \mathcal{V}_t, \mathcal{V}_c, \epsilon_{ma}, \epsilon_{at}, \epsilon_{ac}\}$, where $\mathcal{V}_m, \mathcal{V}_a, \mathcal{V}_t, \mathcal{V}_c$ denote the set of Mashups, APIs, Tags and Categories respectively. In the Web service network $SNet$, if Mashup m invokes API a, then there exists an edge $(m, a) \in \epsilon_{ma}$ between m and a. An edge $(a, t) \in \epsilon_{at}$ exists between a and t or an edge $(a, c) \in \epsilon_{ac}$ exists between a and c if API a is associated with tag t or belongs to category c. If only the interactive information between

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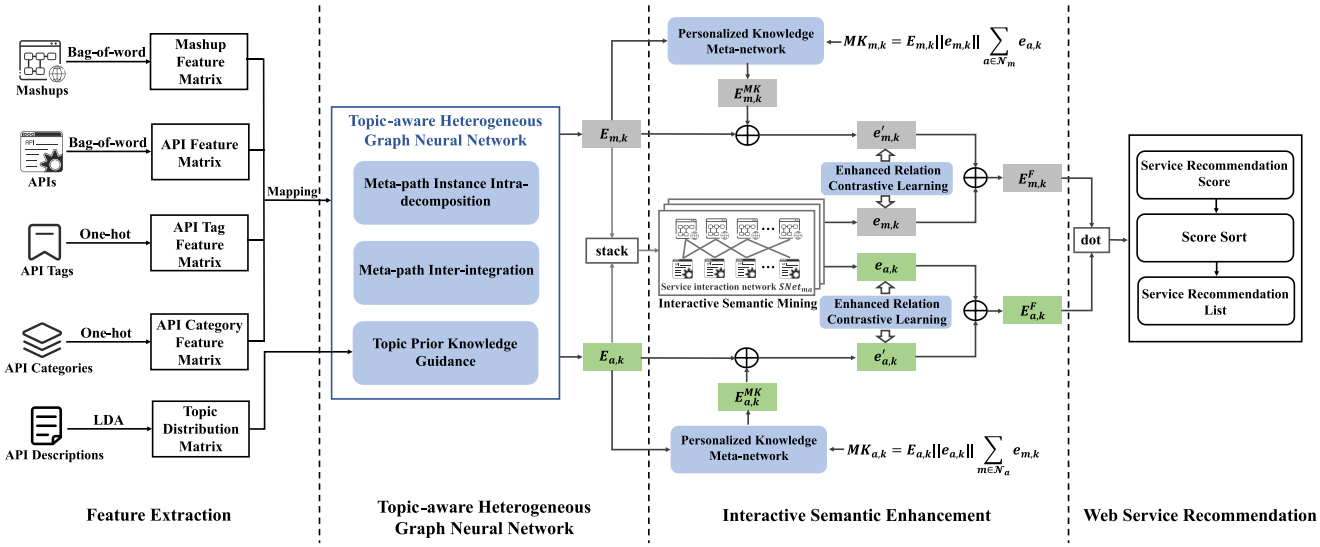

Fig. 2. The overall framework of the proposed method.

Mashups and APIs is retained in the Web service network, it is called the Web service interaction network, formalized as $SNet_{ma} = \{\mathcal{V}_m, \mathcal{V}_a, \epsilon_{ma}\}$.

Definition 2. Meta-path Instance

In the Web service network, meta-path mp consists of a sequence of node types and relationship types, denoted as $\mathcal{V}_1 \xrightarrow{R_1} \mathcal{V}_2 \xrightarrow{R_2} \ldots \xrightarrow{R_l} \mathcal{V}_{l+1}$, which defines a composite relation $R = R_1 \circ R_2 \circ \ldots \circ R_l$ between $\mathcal{V}_1$ and $\mathcal{V}_{l+1}$, where $\circ$ represents a composite operator on the relation. Given a predefined meta-path mp_i for Web service network, a meta-path instance MI_m with Mashup m as the target service node is defined as a sequence of nodes that match the meta-path mp_i node types and for which there are actual links. For each predefined meta-paths, there exists a set of meta-path instances with Mashup m as the target service node, denoted as $MI_m^{mp_i}$, where $mp_i \in MP$, i is the predefined meta-paths index, MP is the set of the meta-paths, i.e., $MP = \{mp_1, mp_2, \ldots, mp_I\}$. I is the number of predefined meta-paths.

Definition 3. Meta-path Instance-based Context

Given a predefined meta-path mp_i and meta-path instance MI_m in Web service network, the meta-path instance-based context of the target service node Mashup m is to remove the target service node m from the sequence of nodes of the meta-path instance, denoted as $C_m = MI_m \backslash \{m\}$. In C_m, the node associated with the API description text is denoted as C_m^{text}. For the set of meta-path instances $MI_m^{mp_i}$ with Mashup m as the target service node, there exists a set of meta-path instances $MI_m^{mp_i}$-based context, denoted as $C_m^{mp_i}$.

IV. Method

The overall framework of the method in this paper is shown in Fig. 2, which includes four parts, i.e., Feature Extraction, Topic-aware Heterogeneous Graph Neural Network, Interactive Semantic Enhancement, and Web Service Recommendation. For better identification, notations used in THISE are summarized in Table I.

TABLE I
Notations Used in the Proposed Models

Notations	Description
MP	The set of the meta-paths, $MP = \{mp_1, mp_2, \ldots, mp_I\}$
I	The number of predefined meta-paths
$MI_m^{mp_i}$	The set of meta-path instances with Mashup m
C_m	The meta-path instance-based context of the Mashup m
K	The total number of topics
λ	The distribution matrix of the topics
$x_{\mathcal{V}}$	The initial feature vector
$\varphi(\mathcal{V})$	The type of node $\mathcal{V}$
ρ	The topic similarity score
$\lvert B \rvert$	The size of batch
β^{mp_i}	The meta-path mp_i attention weight coefficient
$E_{m,k}$	The multifaceted topic-aware representation of Mashup m
$E_{a,k}$	The multifaceted topic-aware representation of API a
$MK_{m,k}$	The personalized meta-knowledge of Mashup m
$MK_{a,k}$	The personalized meta-knowledge of API a
W^{MK}	The personalized parameter transformation matrices
$E_{m,k}^F, E_{a,k}^F$	The final representation of Mashup m and API a

A. Feature Extraction

The initial data of the Web services crawled from the ProgrammableWeb platform is preprocessed to gain the available information such as Mashups, APIs, Categories, Tags, Description, and the relationship between them. For the description document information of APIs and Mashups, the initial feature vectors of APIs and Mashups are first obtained using the Bag-of-Word encoder respectively. Considering that LDA can provide an ideal choice to balance performance and complexity under the current service dataset, it is used to obtain the prior topic distributions of APIs from the description documents of

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APIs. Due to the small number of tags and categories, which are 362 and 164 respectively, the initial feature vectors of tags and categories are obtained directly by the One-Hot encoder.

The description document information of the APIs and the number of topics is used as inputs to the topic model LDA to get the topic distribution of the APIs. As shown in (1).

$$\lambda = \frac{n_{d_k} + \alpha}{N(n_d) + \alpha K} \tag{1}$$

Where n_{d_k} is the number of words belonging to topic k in the description document d, $N(n_d)$ is the total number of words in the description document d, K is the total number of topics, α is the Dirichlet prior parameter, $\lambda \in \mathbb{R}^{N(API)\times K}$ is the distribution matrix of the topics extracted from the description document of APIs, and $N(API)$ is the number of APIs.

Due to the varying information, attributes, and interaction modes carried by different types of nodes, different types of nodes and their feature vectors tend to exist in completely different feature vector spaces, which means that their feature representations in the space will also be different. Therefore, before feature vector learning, it is necessary to map the feature vectors of the nodes located in different vector spaces into the shared potential vector subspace so that the feature spaces of the Mashups, APIs, tags, and categories nodes are aligned into the same subspace, and we name this operation as multifaceted mapping. Assuming that there are k potential topic-aware subspaces, the initial node feature vectors in the Web service network are mapped to the shared k potential topic-aware subspaces using k type-specific linear transformations. The multifaceted mapping is shown in (2).

$$E_{\mathcal{V},k} = P_k^{\varphi(\mathcal{V})} \cdot x_{\mathcal{V}} \tag{2}$$

Where $k = 1, 2, \ldots, K$, and $x_{\mathcal{V}} = \{x_m, x_a, x_t, x_c\}$ is the initial feature vector of the four types of nodes in the Web service network,$E_{\mathcal{V},k} = \{E_{m,k}, E_{a,k}, E_{t,k}, E_{c,k}\}$ is the feature representation after mapping of the initial feature vectors in the kth potential topic-aware subspace. $P_k^{\varphi(\mathcal{V})} \in \mathbb{R}^{\frac{D}{K}\times d_{\varphi(\mathcal{V})}}$ is the kth linear transform weight matrix for nodes of type $\varphi(\mathcal{V})$.

B. Topic-Aware Heterogeneous Graph Neural Network

After feature extraction, the topic-aware heterogeneous graph neural network is used to learn the multifaceted representation of the target service node. The framework of THGNN is shown in Fig. 3, which consists of three parts, i.e., Meta-path Instance Intra-decomposition, Meta-path Inter-integration, and Topic Prior Knowledge Guidance. Meta-path instance intra-decomposition aims to mine the topic distribution in the meta-path instance-based context, thereby decomposing more fine-grained topic semantics for the target service node. Meta-path inter-integration performs specific aggregation of topic-aware semantics based on the inferred topic distribution and uses the multi-facet attention mechanism to integrate different meta-paths. Iterative and alternate execution of meta-path instance intra-decomposition and meta-path inter-integration can better capture the structural semantics and topic semantics of the target

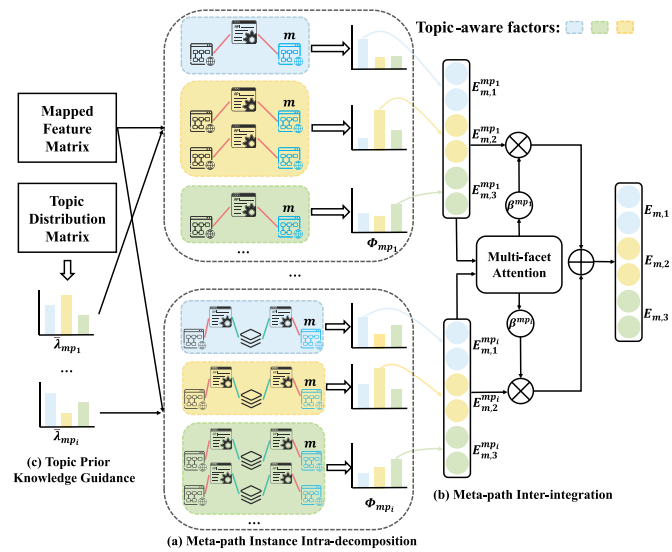

Fig. 3. The framework of THGNN.

service node, thereby learning multifaceted topic-aware representation of the target service node. Topic prior knowledge guidance effectively improves the quality of learning multifaceted topic-aware representations by incorporating topic prior knowledge into meta-path instances context aggregation process.

1) Meta-Path Instance Intra-Decomposition: Given the meta-path mp_i and the set of meta-path instances $MI_m^{mp_i}$-based context $C_m^{mp_i}$, the meta-path instance intra-decomposition first utilizes the context encoder of the meta-path instance to obtain a multifaceted topic-aware representation based on the context of the meta-path instance, and then determines the topic distribution of the context of each of the sampled meta-path instances by calculating the similarity between the multifaceted topic-aware representation of the context and the multifaceted topic-aware representation of the target service node, and thus, can infer to which topic each meta-path instance belongs. The context encoder for meta-path instances obtains multifaceted topic-aware representation of the target service node based on the context of the current meta-path instance by aggregating the meta-path instance-based context information, the context encoder is shown in (3).

$$E_{m,k}^{C_m} = f\left(E_{v,k}, \forall v \in C_m\right) \tag{3}$$

Where $C_m \in C_m^{mp_i}$, $E_{m,k}^{C_m}$ denotes the representation of the context C_m of the target service node m based on the meta-path instance in the kth topic-aware subspace, and the aggregation function $f(\cdot)$ is the average pooling operation.

After encoding the context, the topic similarity score is obtained by computing the cosine similarity between the multifaceted representation of the target service node m and its meta-path instance-based context multifaceted representation. According to the topic similarity score, the most relevant topic-aware subspace shared by the target service node representation and its meta-path instance-based context multifaceted representation can be determined. The topic similarity score is calculated

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as shown in (4).

$$\rho_{k|C_m} = \frac{exp\left(cos\left(E_{m,k},\ E_{m,k}^{C_m}\right)\right)}{\sum_{k'}^{K} exp\left(cos\left(E_{m,k'},\ E_{m,k'}^{C_m}\right)\right)} \tag{4}$$

Where $k = 1, 2, \ldots, K, C_m \in C_m^{mp_i}$. If the target service nodes and its meta-path instance-based context are more relevant in the kth topic-aware subspace, then their k^{th} topic similarity scores will be correspondingly higher. Meanwhile, the topic similarity score can also be regarded as the importance weighting coefficient of meta-path instance-based context propagation to the topic-aware subspace where the target service node is located. By propagating the meta-path instance-based context to the target service node, the multifaceted topic-aware representation of the target service node m specific to the meta-path mp_i can be obtained, as shown in (5).

$$E_{m,k}^{mp_i} = L2_Norm\left(\sum_{C_m \in C_m^{mp_i}} \rho_{k|C_m} \cdot E_{m,k}^{C_m}\right) \tag{5}$$

Where $k = 1, 2, \ldots, K$ and $mp_i \in MP$. In short, the inputs of the meta-path instance intra-decomposition are the initial feature vector $E_{\mathcal{V},k}$ and the set of predefined meta-paths MP, and its outputs are I groups of meta-path-specific multifaceted topic-aware representations for the target service node m.

2) Meta-Path Inter-Integration: Due to the heterogeneous characteristic of Web service network, each service node can present different structural semantics through different meta-paths. Each service node may have different personalized characteristics on different meta-paths, and different meta-paths may also have different importance in capturing structural semantics. If weights are uniformly assigned to each meta-path, they cannot be flexibly adapted to the personalized characteristics of different service nodes, and it is unable to differentiate the contribution of different meta-paths, which leads to the neglect of certain key structural semantics. Different from the common attention mechanism with a single similarity weighting, the multi-facet attention mechanism calculates and weights attention separately in multiple dimensions. Therefore, meta-path inter-integration uses the multi-facet attention mechanism to obtain the importance of predefined meta-paths, and integrates the topic-aware semantics and structural semantics of the target service node to generate the multifaceted topic-aware representation for the target service node that preserves both topic-aware semantics and structural semantics.

To learn the importance weight coefficients of a specific meta-path mp_i, we transform the single-facet representation $E_{m,k}^{mp_i}$ of a specific meta-path mp_i using the weight matrix W, then averaging the representations in the batch to represent the meta-path of the kth topic-aware subspace. It is shown in (6).

$$s_k^{mp_i} = \frac{1}{|B|} \sum_{m \in B} tanh\left(W \cdot \sigma\left(E_{m,k}^{mp_i}\right)\right) \tag{6}$$

Where $k = 1, 2, \ldots, K, \sigma(\cdot)$ is the *ReLU* activation function, B is a batch of training data, $|B|$ is the size of a batch.

Since the multifaceted query vectors $(q_1, q_2, \ldots, q_K)$ have the role of guiding and controlling the allocation of meta-paths attention, the relative importance of the meta-path mp_i in each topic-aware subspace can be determined by calculating the dot-product similarity between the multifaceted query vectors and the single-facet transformation vectors $s_k^{mp_i}$. Because predefined meta-paths describe the specific semantics behind the Web service network, the overall importance of predefined meta-paths should be shared across all topic-aware subspaces. For this, the total attention scores $\rceil^{\Updownarrow}\surd^{)}$ of the predefined meta-paths can be computed by summing the attention scores of each topic-aware subspace, and then adjusting the weights of the attention scores of the different meta-paths by the SoftMax operation. It is shown in (7) and (8).

$$e^{mp_i} = \sum_{k=1}^{K} q_k \cdot s_k^{mp_i} \tag{7}$$

$$\beta^{mp_i} = \frac{exp\left(\rceil^{\Updownarrow}\surd^{)}\right)}{\sum_{mp_j \in MP} exp\left(\rceil^{\Updownarrow}\surd^{|}\right)} \tag{8}$$

Where $q_k \in \mathbb{R}^{\frac{D}{K}}$. $\rceil^{\Updownarrow}\surd^{)}$ is the total attention score of the meta-path mp_i. β^{mp_i} denotes the specific meta-path mp_i attention weight coefficients normalized to the total attention scores of all meta-paths.

After obtaining the attention weight coefficients of all predefined meta-paths $(\beta^{mp_1}, \beta^{mp_2}, \ldots, \beta^{mp_I})$, the multifaceted topic-aware representations of each specific meta-path of the target service node can be integrated to obtain the final multifaceted topic-aware representation of the target service node. It is shown in (9).

$$E_{m,k} = \sum_{mp_i \in MP} \beta^{mp_i} \cdot E_{m,k}^{mp_i} \tag{9}$$

Where $k = 1, 2, \ldots, K$, $E_{m,k}$ is the multifaceted topic-aware representation of the target service node Mashup m combining structural semantics and topic-aware semantics.

The two steps of meta-path instance intra-decomposition and meta-path inter-integration are executed alternately and as a single iteration, and the output $E_{m,k}$ after each iteration will in turn guide the topic inference in the meta-path instance intra-decomposition step in the next iteration. This feedback mechanism allows each iteration to more accurately infer topic information within meta-path instances based on the results of the previous iteration. At the end of the iteration, the final inferred topic-aware distribution matrix of all target service nodes will also be obtained, denoted as $\{\Phi_{mp_i}, mp_i \in MP\}$, where $\Phi_{mp_i} \in \mathbb{R}^{|C_m^{mp_i}| \times K}$. This topic-aware distribution will reflect the importance and degree of association of target service nodes on different topic subspaces, thus providing a comprehensive topic-aware view.

3) Topic Prior Knowledge Guidance: Although the inferred topic-aware distributions are continuously optimized during the iterative process, two issues need to be noted. a) Due to the heterogeneous characteristics of Web service network, the topic-aware subspaces inferred from the meta-path instance

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intra-decomposition step may still be confusing. This confusion may lead to extreme cases of topic-aware subspace allocation, where certain topic-aware subspaces are over-weighted while others are ignored. b) The process of inferring topic-aware distributions can be viewed as a local clustering, where the K potential topic-aware subspaces assumed in inferring topic-aware distributions may not capture global knowledge due to the limitations of local clustering. This paper introduces the topic prior knowledge guidance module to effectively solve the above problems. For issue a), the topic prior knowledge guidance module makes each topic-aware subspace more orthogonal to each other, thus ensuring that each topic subspace has the appropriate size and enhancing the independence between them. For issue b), the topic prior knowledge guidance module guides the inference process of topic-aware distributions by incorporating prior topic information, and such guidance can help to overcome the limitations of local clustering, thus better capturing global knowledge.

Inspired by the orthogonality loss term for graph clustering [40], The topic prior knowledge guidance module acts as a regularization term that uses the topic model LDA to analyze the description text of APIs and extracts the topic prior knowledge from it, which is then used to guide the context aggregation process of Web service network. the regularization term is shown in (10) and (11).

$$\bar{\lambda}_{mp_i} = \frac{1}{|C_m^{mp_i}|} \sum_{c \in C_m^{mp_i}} \lambda_{d_1}^c \tag{10}$$

$$\mathcal{L}_T = \frac{1}{|MP|} \sum_{mp_i \in MP} \left|\left| \frac{\Phi_{mp_i}^T \Phi_{mp_i}}{||\Phi_{mp_i}^T \Phi_{mp_i}||_F} - \frac{\bar{\lambda}_{mp_i}}{||\bar{\lambda}_{mp_i}||_F} \right|\right|_F \tag{11}$$

Where $\lambda_{d_1}^c$ is the topic distribution of the first API node with textual content in the meta-path instance-based context computed by Eq (1). $\bar{\lambda}_{mp_i}$ is the topic prior knowledge, which is used to guide the context aggregation process of the service network. Φ_{mp_i} is the final inferred topic-aware distribution matrix of all target service nodes.

C. Interactive Semantic Enhancement

After encoding through the topic-aware heterogeneous graph neural network, the target service node Mashup m and API a are obtained with the combined topic semantics and structural semantics of the representations $\{E_{m,k},\ E_{a,k} | \ k = 1, 2, \ldots, K\}$. However, meta-paths have the following drawbacks in representing the structural semantics of Web service network. a) Meta-paths usually focus on specific path patterns, so it has insufficient understanding of the interaction perception of the entire Web service network and cannot capture the complex interaction patterns between services. b) Meta-paths introduce indirection, which leads to information missing and blurring when representing the interactivity between Mashups and APIs. To address the above drawbacks, this paper designs the interactive semantic enhancement module, which consists of three parts, i.e., Service Interaction-aware Semantic Mining, Personalized Knowledge Meta-network, and Enhanced Relation Contrastive Learning. Service Interaction-aware Semantic Mining aims to distill the interaction semantics of Mashups and APIs from the Web service interaction network and capture complex interaction patterns between services. Personalized Knowledge Meta-network captures personalized meta-knowledge from the structural semantic and topic-aware semantic side information of Mashups and APIs, which is used to guide the learning of service nodes and adapt them to specific preferences. Enhanced Relation Contrastive Learning aims to enable the transformation of personalized meta-knowledge with adaptive contrast enhancement, which can effectively solve the problem of sparsity of heterogeneous relations and generate more robust feature representations for service nodes.

1) Service Interaction-Aware Semantic Mining: Given the Web service interaction network $SNet_{ma}$, and taking the multifaceted topic-aware representations $\{E_{m,k},\ E_{a,k} \mid k = 1,\ 2,\ \ldots, K\}$ of Mashups and API as inputs, this paper applies graph convolutional neural network as the encoder for the service interaction view, and iteratively refines the interaction information of Mashups and APIs through message propagation mechanism as shown in (12).

$$\begin{cases} e_{m,k}^{l+1} = \sum\limits_{a \in \mathcal{N}_m} \frac{1}{\sqrt{|\mathcal{N}_m|}\sqrt{|\mathcal{N}_a|}} e_{a,k}^l \\ e_{a,k}^{l+1} = \sum\limits_{m \in \mathcal{N}_a} \frac{1}{\sqrt{|\mathcal{N}_a|}\sqrt{|\mathcal{N}_m|}} e_{m,k}^l \end{cases} \tag{12}$$

Where $k = 1, 2, \ldots, K$, $\mathcal{N}_m$ and $\mathcal{N}_a$ are the neighbor node sets of Mashup m and API a in the Web service interaction network respectively. $e_{m,k}^l$ and $e_{a,k}^l$ represent the representation of Mashup m and API a after the lth iteration. The initial input representation $(e_{m,k}^0,\ e_{a,k}^0)$ of the encoder is $(E_{m,k},\ E_{a,k})$. Inspired by the low complexity and excellent generalization capability of Lightweight GCN, the encoder of service interaction view does not set linear transformations and nonlinear activations. Its form is shown in (13).

$$\begin{cases} e_{m,k} = e_{m,k}^0 + \sum\limits_{l=1}^{L} \frac{e_{m,k}^l}{||e_{m,k}^l||} \\ e_{a,k} = e_{a,k}^0 + \sum\limits_{l=1}^{L} \frac{e_{a,k}^l}{||e_{a,k}^l||} \end{cases} \tag{13}$$

Where $k = 1,\ 2, \ldots, K$, L denotes the number of iterations of the GCN layer.

2) Personalized Knowledge Meta-Network: In actual application scenarios of Web services, the impact of Mashup-side information and API-side information on the interaction patterns between Mashups and APIs may vary between different Mashups. For example, each Mashup has its own specific needs and goals, and some Mashup developers choose an API more for how well it matches their functionality, while others are more interested in the compatibility and reliability of the API. Therefore, personalized knowledge transfer of side information is necessary to guide Mashups to learn and adapt to specific preferences. Inspired by graph meta-networks [41], the personalized knowledge meta-network is designed to capture personalized knowledge in side information.

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Firstly, important features in Mashups and APIs need to be preserved in side information and interaction-aware information. This important feature is also called meta-knowledge. The specific operation of capturing meta-knowledge is shown in (14).

$$\begin{cases} MK_{m,k} = E_{m,k} \, ||e_{m,k}|| \sum\limits_{a \in \mathcal{N}_m} e_{a,k} \\ MK_{a,k} = E_{a,k} \, ||e_{a,k}|| \sum\limits_{m \in \mathcal{N}_a} e_{m,k} \end{cases} \tag{14}$$

Where $k = 1, 2, \ldots, K$, $MK_{m,k}$ and $MK_{a,k}$ denote the personalized meta-knowledge extracted from the side information knowledge of Mashup and API, respectively, which includes the structural and topic-aware semantic information, interaction-aware semantic information, and their respective domain information. By comprehensively considering information from these three dimensions, meta-knowledge can effectively reflect important contextual signals for personalized knowledge delivery.

Then, the extracted personalized meta-knowledge is used to generate a personalized parameter transformation matrix, as shown in (15).

$$\begin{cases} f^1_{mlp}\left(MK_{m,k}\right) \rightarrow W^{MK_1}_{m,k} \\ f^2_{mlp}\left(MK_{m,k}\right) \rightarrow W^{MK_2}_{m,k} \end{cases} ; \begin{cases} f^1_{mlp}\left(MK_{a,k}\right) \rightarrow W^{MK_1}_{a,k} \\ f^2_{mlp}\left(MK_{a,k}\right) \rightarrow W^{MK_2}_{a,k} \end{cases} \tag{15}$$

Where $k = 1, 2, \ldots, K$, f^1_{mlp} and f^2_{mlp} are meta-knowledge linear transformation methods, which contain two fully connected layers and have a PReLU activation function. W^{MK_1} and W^{MK_2} are personalized transformation matrices obtained after linear transformation of meta-knowledge to achieve personalized knowledge transfer. This paper limits the rank of the two sets of personalized parameter matrix transformations, which reduces the number of trainable parameters for meta network learning while controlling the risk of overfitting.

Finally, this paper uses the generated personalization parameter matrix and nonlinear mapping function to construct the personalized knowledge meta-network, as shown in (16).

$$\begin{cases} E^{MK}_{m,k} = \sigma\left(W^{MK_1}_{m,k} W^{MK_2}_{m,k} E_{m,k}\right) \\ E^{MK}_{a,k} = \sigma\left(W^{MK_1}_{a,k} W^{MK_2}_{a,k} E_{a,k}\right) \end{cases} \tag{16}$$

Where $k = 1, 2, \ldots, K$, $\sigma(\cdot)$ is the PReLU activation function. $E^{MK}_{m,k}$ and $E^{MK}_{a,k}$ are personalized meta-knowledge extracted from side information knowledge respectively to enhance the interaction-aware semantics between Mashups and APIs. The fusion process of the final representation of Mashups and APIs adopts the weighted summation method, as shown in (17).

$$\begin{cases} E^F_{m,k} = \mu * e_{m,k} + (1-\mu) * e'_{m,k} \\ E^F_{a,k} = \mu * e_{a,k} + (1-\mu) * e'_{a,k} \end{cases} \tag{17}$$

Where $k = 1, 2, \ldots, K$. $e'_{m,k} = E_{m,k} + E^{MK}_{m,k}$ and $e'_{a,k} = E_{a,k} + E^{MK}_{a,k}$. μ is a weight hyperparameter that balances between interaction-aware semantics and structure and topic-aware semantics with personalized meta-knowledge. $E^F_{m,k}$ and $E^F_{a,k}$ denote the final representations of Mashups and APIs for the service recommendation task, respectively, which are augmented with interaction-aware semantics based on the combination of structural semantics and topic-aware semantics.

3) Enhanced Relation Contrastive Learning: To further enhance the representation learning and address the problem of sparsity of Web service network interactions, this paper uses Enhanced Relation Contrastive Learning. By introducing more supervised signals, we can enhance the learning process of service heterogeneous relationships, thus improving the robustness of feature representations. The challenge of heterogeneous relationship sparsity can be effectively addressed by self-enhanced contrastive learning techniques to generate more robust feature representations for each service node.

With the facilitation of the two modules of topic-aware heterogeneous graph neural network and interactive semantic enhancement, two sets of representations $(e_{m,k}, e'_{m,k})$ and $(e_{a,k}, e'_{a,k})$ of Mashup and API can be obtained. Inspired by the positive performance of self-supervised learning in recommendation tasks, this paper employs InfoNCE-based contrastive learning loss to enhance the learning of Mashups representations and APIs representations. The quality and robustness of the learned representations can be improved by applying InfoNCE-based contrastive learning loss between the two sets of representations. The InfoNCE-based contrastive learning loss for Mashup nodes is shown in (18).

$$\mathcal{L}^m_{cl} = -\frac{exp\left(\sum_{k=1}^{K} s\left(e_{m,k},\ e'_{m,k}\right)/\tau\right)}{\sum_{m' \in \mathcal{V}_m} exp\left(\sum_{k=1}^{K} s\left(e_{m',k},\ e'_{m,k}\right)/\tau\right)} \tag{18}$$

Where K is the total number of topics, $s(\cdot)$ is the inner product similarity function, τ is a temperature hyperparameter that regulates the sensitivity of the similarity measure, m' denotes negative samples. Similarly, the InfoNCE-based contrastive learning loss $\mathcal{L}^a_{cl}$ of API nodes can be obtained. Finally, the total contrastive learning loss is $\mathcal{L}_{cl} = \mathcal{L}^m_{cl} + \mathcal{L}^a_{cl}$.

D. Web Service Recommendation

After obtaining the high-quality representations $E^F_{m,k}$ and $E^F_{a,k}$ of Mashups and APIs containing rich semantics, the similarity of the service recommendation pairs $(m \in \mathcal{V}_m, a \in \mathcal{V}_a)$ is evaluated by inner product, which is then summed up to be the final recommendation scores of the Web service recommendations. Its form is shown in (19).

$$r_{ma} = \sum_{k=1}^{K} \left(E^F_{m,k}\right)^T \cdot E^F_{a,k} \tag{19}$$

Where r_{ma} is the recommendation score between Mashup m and API a. Higher r_{ma} can reflect higher interaction probability. In Web service recommendation, we sort the recommendation scores between each Mashup node and all API nodes in the test set, and then select a group of APIs with high recommendation scores for recommendation.

To optimize the learning of representations through the Web service recommendation task, this paper employs the Bayesian Personalized Ranking (BPR) pair-wise loss function as

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shown in (20).

$$\mathcal{L}_{bpr} = \sum_{(m,a^+,a^-)\in O} -ln\left(sigmoid\left(r_{ma^+} - r_{ma^-}\right)\right) \quad (20)$$

Where $O \in \{O^+ \cup O^-\}$. For Mashup m, a positive sample API a^+ that has interacted with m belongs to O^+, while a negative sample API a^- that has not interacted with m belongs to O^-. Combining the BPR loss function $\mathcal{L}_{bpr}$, the topic prior regularization term $\mathcal{L}_T$, and the InfoNCE-based contrastive learning loss $\mathcal{L}_{cl}$, the overall training loss can be obtained as shown in (21).

$$\mathcal{L} = \mathcal{L}_{bpr} + \zeta * \mathcal{L}_T + \xi * \mathcal{L}_{cl} + \gamma ||\Theta||^2 \quad (21)$$

Where ζ is the weight hyperparameter of the topic prior regularization term. ξ is the coefficient of InfoNCE-based contrastive learning loss. γ is the coefficient of the regularization term. Θ is the weight decay regularization term.

V. Experiments

In this section, extensive service recommendation experiments on real-world datasets are performed to validate the effectiveness of the method proposed in this paper. These experiments are designed to address the following research questions:

- **Q1:** How does the performance of the proposed method in this paper compare to existing methods in service recommendation?
- **Q2:** Is it beneficial for the key modules in the proposed method to service recommendation?
- **Q3:** How do key hyperparameters affect model performance?

A. Experimental Settings

1) Datasets: Web service metadata crawled from the ProgrammableWeb (http://49.123.3.158/download/ProgrammableWeb2022.zip).The metadata set is divided into two parts, i.e., Mashup and API. The Mashup part has a total of 6466 Mashups. The fields in this part include Mashup name, related API, description document, category, development company, and date. The API part has a total of 22650 APIs. The fields in this part include API name, description document, category, tag, developer, and date. We performed a series of preprocessing operations on the metadata set, such as deduplication, removing isolated nodes, and deleting useless fields, to obtain a ProgrammableWeb subset. Specifically, the Mashup part removed useless fields such as development company and date, and eliminated Mashups with no historical interactions, categories, and description documents. The API part deleted useless fields such as developers and development dates. And only keep APIs that have historical interactions with Mashup and have categories and description documents. This subset includes 5749 Mashups and their corresponding description documents, 1196 APIs and their corresponding description documents, 362 tags, and 164 categories. The statistics of the dataset used in the experiments are shown in Table II. Based on the preprocessed dataset and according to the interactions between different types of nodes, three adjacency matrices are constructed, namely Mashups-APIs, APIs-Tags, and APIs-Categories. Combining these three adjacency matrices, the Web service network $SNet$ is constructed, which includes four types of nodes, 7471 nodes, and 14112 links. In addition, the detailed description of the predefined meta-paths set is shown in Table III.

TABLE II
Statistics of Experimented Datasets

Relation(X-Y)	#X	#Y	#X-Y
API-Mashup	1196	5749	10579
API-Tag	1196	362	2337
API-Category	1196	164	1196

TABLE III
Description of Predefined Meta-paths

Notation	Meta-path	Semantics
MAM	Mashup-API-Mashup	Mashup calling the same API
MATAM	Mashup-API-Tag-API-Mashup	Mashup calling the API that is under the same tag
MACAM	Mashup-API-Category-API-Mashup	Mashup calling the API that is under the same category

2) Baselines: To evaluate the validity of the proposed method THISE, it is compared with three categories of graph neural network methods, i.e., homogeneous graph neural network-based (GraphSAGE, DisenGCN, LightGCN), heterogeneous graph neural network-based (HAN, HGT, THGNN), graph self-supervised learning-based (Heco, SMIN, HGCL).

- *GraphSAGE* [42]: It is a classical GNN method for encoding homogeneous graph using samplers and aggregators.
- *DisenGCN* [43]: It is a GNN method for homogeneous graph disentanglement, which utilizes domain routing mechanisms and embedding propagation to disentangle potential factors between nodes and their neighbors.
- *LightGCN* [44]: It is a simple, linear, and neat graph convolutional neural network (GCN) method, which simplifies the design of GCN and pays more attention to the process of neighborhood aggregation.
- *HAN* [13]: It is a representative HGNN method, which utilizes a two-layer attention mechanism to fully consider the importance of node paths and meta-paths, and generates node embeddings by hierarchically aggregating the features of meta-path-based neighbors.
- *HGT* [45]: It designs parameters related to node and edge types to characterize heterogeneous attention on each edge to refine user and item embeddings as well as various relationships in the heterogeneous graph structure.
- *THGNN* [14]: It hierarchically mines potential topic-aware semantics, adopts two-step aggregation mechanism, and uniquely aggregates rich heterogeneous structures based on inferred topic-aware factors.
- *Heco* [46]: It combines contrast learning mechanism with heterogeneous graph neural network to capture both local

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TABLE IV
EXPERIMENTAL RESULTS OF SERVICE RECOMMENDATION

Model	HR@10	HR@15	HR@20	HR@25	HR@30	NDCG@10	NDCG@15	NDCG@20	NDCG@25	NDCG@30
GraphSAGE	0.3996	0.4117	0.4499	0.4684	0.5524	0.2109	0.2325	0.2692	0.2746	0.3020
DisenGCN	0.4331	0.4534	0.4826	0.5226	0.5540	0.3070	0.3138	0.3260	0.3454	0.3474
LightGCN	0.3769	0.4098	0.4431	0.4639	0.4802	0.2382	0.2458	0.2526	0.2574	0.2607
HAN	0.4907	0.5085	0.5414	0.5671	0.5811	0.3282	0.3324	0.3383	0.3424	0.3456
HGT	0.5059	0.5444	0.5754	0.5902	0.6007	0.3421	0.3534	0.3563	0.3609	0.3651
THGNN	0.6200	0.6609	0.6866	0.7081	0.7319	0.4511	0.4619	0.4679	0.4726	0.4775
Heco	0.5584	0.5837	0.6189	0.6299	0.6461	0.4448	0.4516	0.4600	0.4624	0.4658
SMIN	0.5632	0.5913	0.6246	0.6510	0.6669	0.4646	0.4736	0.4820	0.4878	0.4912
HGCL	0.6318	0.6843	0.7248	0.7505	0.7679	0.4579	0.4719	0.4815	0.4871	0.4907
THISE	**0.6820**	**0.7146**	**0.7376**	**0.7621**	**0.7762**	**0.5105**	**0.5192**	**0.5246**	**0.5299**	**0.5329**

and higher-order structures by learning network patterns and node embeddings of meta-path views.

- *SMIN* [47]: It is a self-supervised meta-graph information network that integrates social and knowledge-aware relational structures into a user preference framework.
- *HGCL* [48]: It is a heterogeneous graph contrastive learning approach that incorporates the semantics of heterogeneous relationships into the modeling of user-item interactions and enhances knowledge transfer through contrastive learning between different views.

3) Parameter Settings: THISE model is implemented by using PyTorch. The Adam optimizer is used to learn parameters of the model, and the learning rate is set to 1e-7 $\sim$ 1e-4. During the model implementation process, the number of API potential topic-aware factors is selected from $\{2, 4, 8, 16, 32\}$, the number of graph convolution layers is selected from $\{1, 2, 3, 4, 5\}$, and the embedding size is tuned from the range of $\{8, 16, 32, 64, 128, 256\}$. Additionally, 25% of the interactions between Mashups and APIs are randomly hidden as the positive test samples for evaluating the accuracy of service recommendations, and the rest of the non-interaction samples are used as the negative test samples. In the training stage, the model is trained by using the remaining 75% of the interactions between Mashups and APIs as positive training samples, and the same number and form of non-interacting samples are randomly selected as negative samples. Such training and testing settings ensure that the THISE model is effectively trained and evaluated in the service recommendation task. To measure service recommendation accuracy and satisfaction of different methods, two widely used recommendation evaluation metrics are used, i.e., Hit Ratio (HR) and Normalized Discounted Cumulative Gain (NDCG), with (22) and (23).

$$HR = \frac{1}{N} \sum_{i=1}^{N} hit\,(i) \tag{22}$$

$$NDCG = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{log_2\,(p\,(i) + 1)} \tag{23}$$

Where N is the number of test samples. $hit(i)$ is used to represent whether the ith demand item is included in the recommended list of items. $p(i)$ is the position of the ith demand item in the recommended list of items.

B. Performance Comparison (Q1)

The service recommendation performance of all compared methods on the ProgrammableWeb dataset is shown in Table IV and Fig. 4. High HR values indicate high accuracy of service recommendations and also show that the recommendation list accurately meets the interests and needs of the users. High NDCG values indicate high user satisfaction with the relevance and rank order of the recommendation results, reflecting high-quality service recommendations. From the experimental results, the following key observations are summarized:

- THISE outperforms all the baseline methods in service recommendation, which shows the effectiveness of THISE. These performance improvements are attributed to the following aspects: (a) Through meta-path instance intra-decomposition, THISE identifies the multifaceted topic-aware factors behind the connections of associated nodes in the service network. (b) THISE fully utilizes the heterogeneous structure and unstructured text content in the service network and combines them effectively. (c) The prior topic distribution as global knowledge can effectively guide the inferencing of topic-aware factors and the aggregation of context information. (d) THISE can refine the interaction semantics of Mashups and APIs and capture complex interaction patterns between services. (e) Adaptive contrastive learning can improve service recommendation performance via using self-supervised signaling between service heterogeneous relationships.
- In homogeneous graph neural network-based methods, LightGCN is relatively lightweight and focuses more on simplified fully-connected neighbor propagation, which has some limitations on its capability to model complex service network structures, resulting in poor service recommendation. GraphSAGE samples and aggregates the neighbors of service nodes in the service network, but the lack of global optimization leads to limitations in the representation of service nodes. DisenGCN can dynamically identify and disentangle the potential factors of existing edges through the neighbor routing mechanism. Therefore, among similar GNN methods, DisenGCN performs best.
- The heterogeneous graph neural network-based methods perform better than the homogeneous graph neural network-based methods on the service recommendation task, which further demonstrates the effectiveness of

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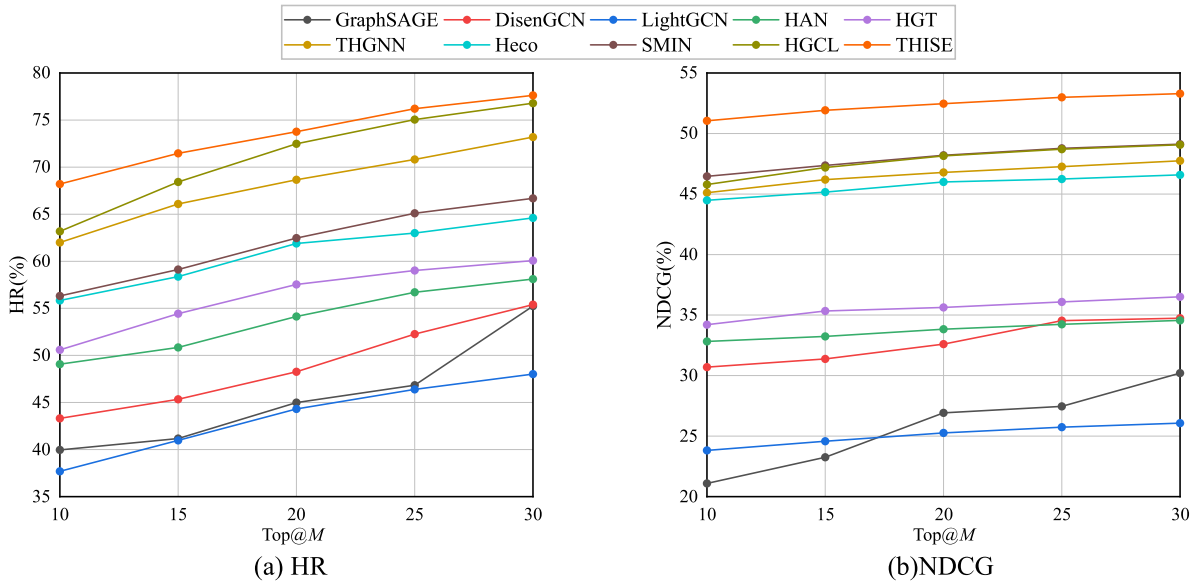

Fig. 4. Service recommendation performance comparison of different methods.

considering heterogeneous relational knowledge in the recommender system. Compared to the two-layer conventional attention mechanism of HAN, HGT performs adaptive attention allocation based on the importance of different types of nodes. This mechanism can more flexibly adapt to the complexity of the service network and show better performance in service recommendation. THGNN far outperforms other heterogeneous graph neural network-based methods in service recommendation accuracy and satisfaction, which is attributed to that THGNN not only takes into account heterogeneous structural features in the service network, but also identifies fine-grained topic-aware semantics from the textual content of the nodes. These observations also confirm the necessity and benefits of mining fine-grained topic-aware semantics in the service network.

- The graph self-supervised learning-based methods significantly outperform the other baseline methods in the service recommendation results. Among them, Heco captures the structural similarity and global structural information between nodes through comparative learning, but it has drawbacks in capturing users' personalized preferences. SMIN builds a comprehensive dependency representation for users and APIs by integrating the service network structure into user preferences, but its high dependence on service interactions limits performance in sparse interaction scenarios. In comparison, HGCL shows better recommendation performance, due to its meta-network contrast learning and adaptive contrast enhancement mechanism. HGCL can flexibly adjust the knowledge conversion process based on the unique characteristics of users and APIs, and accurately capture user preferences and API characteristics, which further emphasizes the important role of personalized knowledge transfer in service recommendation.

TABLE V
ABLATION STUDY ON KEY MODULES OF THISE

Metric	HR @10	HR @20	HR @30	NDCG @10	NDCG @20	NDCG @30
w/o-ma	0.6612	0.7293	0.7614	0.5017	0.5189	0.5258
w/o-tp	0.6703	0.7314	0.7709	0.5055	0.5232	0.5296
w/o-pk	0.6733	0.7318	0.7682	0.5018	0.5184	0.5244
w/o-cl	0.6563	0.7198	0.7603	0.5025	0.5173	0.5245
THISE	**0.6820**	**0.7376**	**0.7762**	**0.5105**	**0.5246**	**0.5329**

C. Ablation Study (Q2)

To investigate the validity of submodules in the THISE model, four simplified variants are implemented to demonstrate the impact of each submodule design on service recommendations. The variants are specified as follows:

- **w/o-ma:** This variant removes the multi-facet attention mechanism of meta-paths and assigns the same weight to all meta-paths.
- **w/o-tp:** This variant removes the topic prior knowledge guidance module by setting the regularization term coefficient of this module to 0.
- **w/o-pk:** This variant removes the personalized knowledge meta-network by deleting a series of personalized knowledge migration steps.
- **w/o-cl:** This variant disables the contrastive learning by removing the InfoNCE-based contrastive learning loss $\mathcal{L}_{cl}$ from the total training loss $\mathcal{L}$.

The results of the ablation studies are shown in Table V and Fig. 5. The following conclusions can be obtained by observing the results of the ablation studies: 1) THISE outperforms the variant w/o-ma that does not include the multi-facet attention mechanism in the service recommendation task, indicating that the multi-facet attention mechanism plays a positive role in the model's performance. Fig. 6 illustrates the attention scores for

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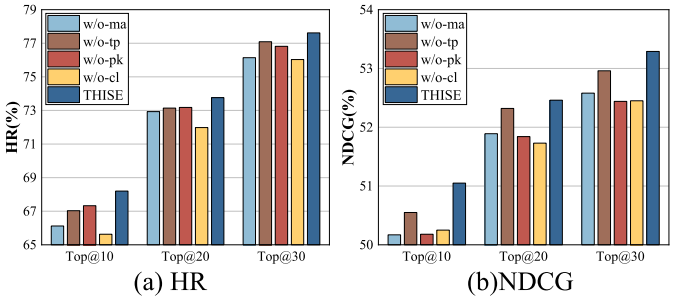

Fig. 5. Impact of key modules in THISE.

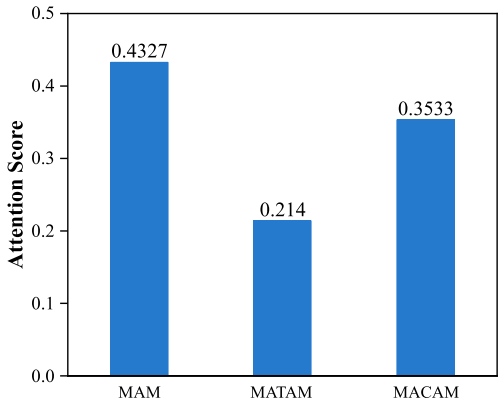

Fig. 6. Attention scores of predefined meta-paths.

predefined meta-paths, which further shows the effectiveness of the attention mechanism in selecting and weighting different meta-paths. 2) The variant w/o-tp performs slightly lower than THISE, which reveals the validity of the topic prior knowledge guidance module for service recommendation. By multiplying the transposed feature matrix with the original feature matrix, Fig. 7 visualizes the correlation analysis between the multi-faceted node embedding elements learned by the service recommendation task. Fig. 7(a), (b), and (c) show the heat map of the degree of correlation between the node embedding elements of the HGCL, the variant w/o-tp, and THISE, respectively, where the number of topic-aware factors K is set to 4, and the embedding size is set to 256. As shown in Fig. 7(a), the features learned by HGCL are still hiddenly entangled. As can be seen in Fig. 7(c), THISE can learn more independent multifaceted representations because the figure shows K clear diagonal blocks. Comparing Fig. 7(b) and (c), it can be further concluded that the topic prior knowledge guidance module plays an important role in providing more orthogonality and maintaining the quality of multifaceted topic-aware embedding. 3) w/o-pk performs worse than THISE on service recommendation, which indicates that personalized knowledge transfer of specific preferences for services helps to learn the representation of services. 4) In all cases, THISE outperforms w/o-cl, reflecting the fact that enhanced relation contrastive learning can effectively alleviate the problem of heterogeneous relationship sparsity in service networks.

D. Hyperparameter Analysis (Q3)

Parameter sensitivity analysis is further performed. Lower Top numbers may not be able to fully evaluate the performance of service recommendations, and higher Top numbers may be affected by more noise and randomness. To achieve a tradeoff, Top@20 is chosen as the number of recommended services in the hyperparametric experiment to evaluate the performance of service recommendation. The experimental results of the hyperparameter sensitivity analysis are shown in Fig. 8. Following conclusions can be drawn from the experiment results:

- **The number of topic-aware factors K.** The number of topic-aware factors is selected from {2, 4, 6, 8, 16, 32}. It can be noticed that as the number of topic-aware factors increases, the performance begins to improve, but when the number of topic-aware factors exceeds 4, the performance drops significantly. This may be because the service network is relatively sparse and adding more topic-aware factors introduces more noise and inaccuracy.
- **The number of GCN propagation layers L.** The number of GCN propagation layers is selected from 1 to 5. It can be observed through the curves that two propagation layers lead to better performance of the model. An appropriate number of layers can capture higher-order neighbor and semantic information, but deeper GCN structures may lead to over-smoothing of the model and can introduce noise to the feature representation.
- **The hidden layer dimension d.** The hidden layer dimension is selected from {8, 16, 32, 64, 128, 256}. As the dimension increases, the performance of the model first improves and then saturates at a dimension size of 64. Hence, properly enlarging the embedding dimension can boost the recommendation performance, but not always bring performance gain due to model overfitting.
- **The topic prior regularization term coefficient ζ.** Service recommendation works best when the topic prior regularization term coefficient is 0.1. When the coefficient is greater than 0.1, the performance decreases rapidly. This may be due to overly sharp topic-aware distribution inference that fails to maintain the correlation between topics, thus negatively affecting the model.
- **The balancing semantic weight coefficient μ.** Service recommendation performs best when the balancing semantic weighting coefficient is 0.8. This shows that the model will place more weight on the interaction semantics of services, which further proves the necessity of enhancing service interaction semantics.
- **The contrastive learning loss coefficient ξ.** Service recommendation performs best when the contrastive learning loss coefficient is 0.32. Contrastive learning loss coefficients that are too low can make the model more reliant on other loss functions and fail to fully incorporate the learning signals from contrast learning. Excessively high contrastive learning loss coefficients may cause the model to focus too much on the similarity between samples, while ignoring wider context information.

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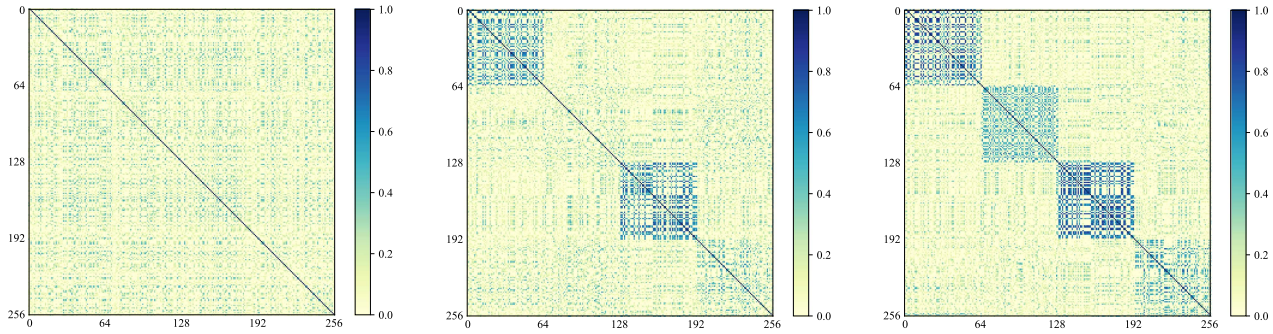

(a) HGCL (b) w/o-tp (c) THISE

Fig. 7. Relevance degree between embedding elements.

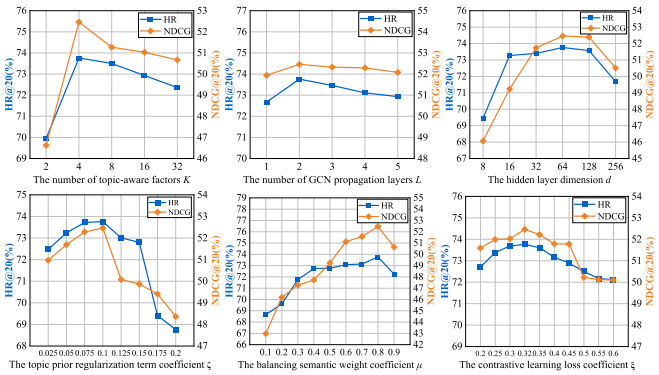

Fig. 8. Hyperparameter study of the THISE.

VI. Conclusion and Future Work

A Web service recommendation method via combining topic-aware heterogeneous graph representation and interactive semantic enhancement is presented in this paper, aiming to improve the accuracy of service recommendation and user satisfaction. First of all, it adopts the two-layer aggregation mechanism of meta-path instance intra-decomposition and meta-path inter-integration. Multifaceted topic-aware factors are first inferred from the meta-path instance-based context using meta-path instance intra-decomposition. Then, the topic-aware semantics and structural semantics of different meta-paths are integrated using meta-path inter-integration, so that the service node generates the multifaceted topic-aware representation that preserves the topic-aware semantics and structural semantics. Additionally, it introduces the topic prior knowledge guidance module to guide the inference of topic factors in the right direction. Finally, it designs the interactive semantic enhancement module to address the missing and blurring of service interaction semantic information caused by meta-paths. The comparative experiments performed on the ProgrammableWeb dataset demonstrate the effectiveness of the proposed method and show that the proposed method significantly improves the accuracy and satisfaction of Web API recommendations. However, the sparse service network caused by less available service data is a potential drawback and limitation of the proposed method. This may have a certain impact on the performance of service recommendation, because a sparse service network will limit the model's accurate understanding and inference of the relationships between services. In future work, we plan to use the Large Language Model (LLM) to deeply mine and understand service description documents and enrich service side information, so as to provide users with more personalized and high-quality service recommendation.

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